Wenbo Lyu

B.Sc. Graduate

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Curriculum Vitae
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Some stuff about me

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.

Education

2021.8-2025.6 **B.Sc. In Geographic Information Science** — Shaanxi Normal University
Xi'an, Shaanxi

Research Experience

2026.6-2026.11 **Research Assistant** — The Hong Kong Polytechnic University
Hong Kong

2024.8-2026.6 **Research Intern** — The Hong Kong University of Science and Technology (Guangzhou)
Guangzhou, Guangdong

Publications

1. Lyu, W., Dai, S., Song, Y., Zhao, W., Yi, W., Xiao, Y., & Jia, N. (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 1–23. https://doi.org/10.1080/13658816.2026.2687121
2. Lyu, W., Lei, Y., Yi, W., Song, Y., Li, X., Dai, S., Qin, Y., & Zhao, W. (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems*, *127*, 102435. https://doi.org/10.1016/j.compenvurbsys.2026.102435
3. Lv, W., Lei, Y., Liu, F., Yan, J., Song, Y., & Zhao, W. (2025). gdverse: An R package for spatial stratified heterogeneity family. *Transactions in GIS*, *29*(2), e70032. https://doi.org/10.1111/tgis.70032
4. Lv, W., Liu, F., Cai, K., Cao, Y., Deng, M., Liang, W., Yan, J., & Wang, G. (2024). Distinguishing the impacts and gradient effects of climate change and human activities on vegetation cover in the weihe river basin, china. *Journal of Geophysical Research: Biogeosciences*, *129*(10), e2024JG008297. https://doi.org/10.1029/2024jg008297
5. Chen, C., Song, Y., Lv, W., Shemery, A., Hampson, K., Yi, W., Zhong, Y., & Wu, P. (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering*. https://doi.org/10.1111/mice.13489
6. Xiao, Y., Yang, Y., Zhao, T., Wang, W., Lv, W., & Zhao, W. (2026). Contrasting causal pathways of vegetation greening between economically strong and weak towns in china's greater bay area. *GIScience & Remote Sensing*, *63*(1), 2623327. https://doi.org/10.1080/15481603.2026.2623327

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.

Unpublished

Co-First Author (1st) **Quantifying pairwise directional influence between spatial variables: Asymmetry, nonlinearity, and beyond** — Submitted to AAAG, currently under review

Co-First Author (2nd) **Spatial Pattern Entropy: A Symbolic Measure for Continuous Spatial Data** — Submitted to IJGIS, currently under review

First Author **Unveiling interaction between spatial processes with information decomposition** — In writing

First Author	**Pattern Causality for Detecting Interactions with the R Package pc**	In writing
First Author	**coupling: Coupling Coordination Analysis in R**	Plan
First Author	**Spatial causal discovery under latent confounders via deep orthogonal representation learning**	Plan

Developed Spatial Analysis Toolkit

Package	Description	Language
spEDM	Spatial Empirical Dynamic Modeling	C++, R
tEDM	Temporal Empirical Dynamic Modeling	C++, R
infoxtr	Information-Theoretic Measures for Revealing Variable Interactions	C++, R
pc	Pattern Causality Analysis	C++, R
gdverse	Analysis of Spatial Stratified Heterogeneity	R, C++, Python
sdsfun	Spatial Data Science Complementary Features	R, C++
coupling	Coupling Coordination Analysis	C++, R
gobi	General ODE-Based Causal Inference	C++, R
geocomplexity	Mitigate Spatial Bias Through Geographical Complexity	C++, R, C
HSAR	Hierarchical Spatial Autoregressive Model	C++, R
GD	Geographical Detectors for Assessing Spatial Factors	R
geosimilarity	Geographically Optimal Similarity	R